

Instruction

Homework 1 covers Chapter 1 to Chapter 2. Total 100 points.

Due by 8:00 AM Apr. 10th, 2026.

Submit your homework (hand-written or printed) in class.

Grading. Please briefly explain your answer. Answer without explanation will only receive half credit. **Plagiarism is strictly forbidden, and will receive zero credit for the entire homework (30% of this course) section.**

Should you have any question on this homework, you could i) ask in class, ii) post on canvas, or iii) post in class QQ group, and we will answer them as soon as possible.

Questions**1. The Internet (2 points)**

What are the key functions of the two parts (i.e., Edge and Core) of the Internet? What are the key characteristics of their functions?

2. Packet Switching v.s. Circuit Switching (4 points)

Is the end-to-end delay in a packet switching system always smaller than the same system with circuit switching? Why or why not?

In addition, consider latency-sensitive applications such as online gaming or real-time AI services (e.g., LLM-based interactive systems). Which switching technique is more suitable for such applications, and why?

3. Reference Model (6 points)

Which of the layers in the OSI reference model execute the following function?

- a) Providing reliable, connection-oriented path between the source and the destination.
- b) Determining which user may have access to the wireless channel.
- c) Framing.
- d) Determining which interface should an IP datagram go out.

What are the main differences between the OSI reference model and the TCP/IP protocol stack?

4. Delay (8 points)

The distance between the sender and receiver is 1000 Km. The propagation speed of the signal on this medium is 1×10^8 m/s. Find the transmission delay and propagation delay for the following two scenarios:

- a) The length of the data is 2×10^7 bits, and the transmission rate is 100 Kbit/s (100 Kbps).
- b) The length of the data is 2×10^3 bits, and the transmission rate is 1 Gbps. What conclusion can you draw from these results?

5. Physical Layer (4 points)

What are the key issues the Physical Layer deals with, and what are the main characteristics/functionalities of the Physical Layer?

6. Bandwidth, Data rate, and Capacity (12 points)

There is a link of data rate 6 Mbps, to be shared by 30 users. Suppose each user, when active, needs a 500 Kbps data rate. Each user is active with a probability of 0.3.

- a) If circuit switching is used, how many users can be hosted on this link?
- b) If packet switching is used, what is the probability that the link is overloaded? (You only need to formulate the expression; no need to compute the exact value.)
- c) What is the minimum signal-to-noise ratio (SNR) to provide for the required 6 Mbps data rate on a channel of bandwidth 30 MHz?

7. Shannon's Theorem (5 points)

A channel has a signal-to-noise ratio (SNR) of 30 dB and a bandwidth of 2 MHz. According to Shannon's theorem, calculate the maximum data transmission rate for this channel.

8. Multiplexing (4 points)

What are the common channel multiplexing techniques? Why is channel multiplexing necessary?

9. CDMA (4 points)

Consider 4 stations with the following chip sequences.

$$A : (-1 - 1 - 1 + 1 + 1 - 1 + 1 + 1)$$

$$B : (-1 - 1 + 1 - 1 + 1 + 1 + 1 - 1)$$

$$C : (-1 + 1 - 1 + 1 + 1 + 1 - 1 - 1)$$

$$D : (-1 + 1 - 1 - 1 - 1 - 1 + 1 - 1)$$

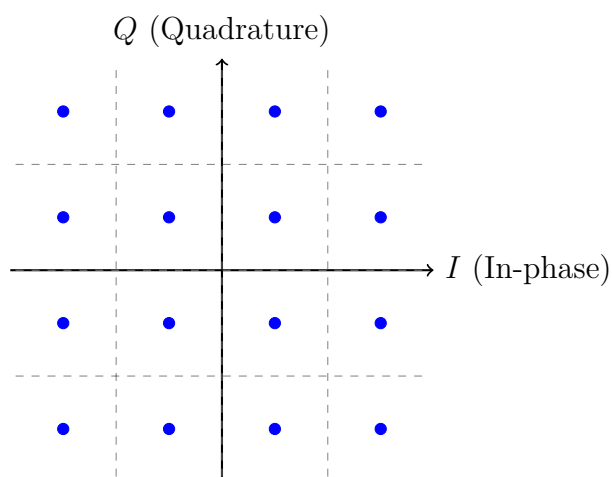
The received sequence $S = (-1 + 1 - 3 + 1 - 1 - 3 + 1 + 1)$.
Which station transmitted, and what are the transmitted bits?

10. Reference Models (10 points)

Briefly explain the main differences between the OSI reference model and the TCP/IP reference model. Why did the TCP/IP model become the dominant standard for the Internet?

11. Modulation and Constellation Diagram (10 points)

The figure below shows a constellation diagram discussed in Chapter 2.



- Based on the provided constellation diagram, what specific modulation technique is being used? Explain your reasoning.
- How many bits are transmitted per symbol in this modulation scheme?

- c) If the symbol rate (baud rate) of the channel is 2000 baud, what is the data transmission rate in bps?

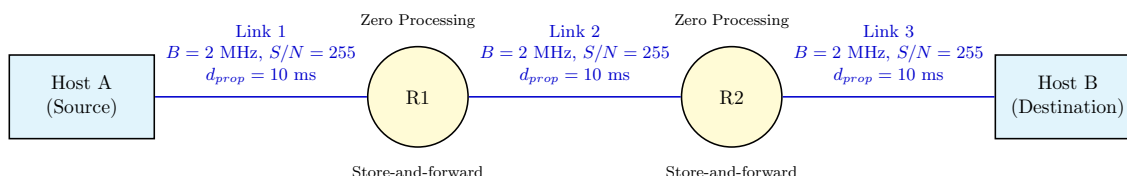
12. Throughput and Bottleneck (10 points)

Consider a path between a server and a client consisting of three links with transmission rates $R_1 = 10$ Mbps, $R_2 = 2$ Mbps, and $R_3 = 5$ Mbps.

- a) What is the maximum throughput for a large file transfer between the server and the client?
- b) Which link is the bottleneck link? Briefly explain why.

13. Store-and-Forward Delay and Channel Capacity (15 points)

The figure below illustrates a packet switching process across a multi-hop network.



Suppose the source host wants to send a file of $L = 3.2 \times 10^7$ bits to the destination host. The path consists of 3 links (i.e., 2 routers in between, similar to the multi-hop process shown in the figure). Instead of a given fixed transmission rate, assume each link operates over a noisy channel with a bandwidth of 2 MHz and a signal-to-noise ratio (S/N) of 255. The propagation delay for each link is $d_{prop} = 10$ ms. Assume there are no processing or queuing delays.

- a) According to Shannon's theorem, calculate the transmission data rate R (in Mbps) for each link.
- b) If the file is sent as a single large packet, calculate the total end-to-end delay from the source to the destination.
- c) If the file is divided into 4 equal-sized packets (ignoring header overhead) and sent sequentially, calculate the total end-to-end delay. Does packetizing the file reduce the total delay? Briefly explain why.

14. Standardization (6 points)

Internet standards are crucial for interoperability.

- a) What is the name of the primary document series used by IETF to publish Internet standards and protocols?
- b) Find the earliest and the most recent RFCs that specifies IPV6, list their current status, and analyze their updating relationship.